CSE400 Gaussian Distribution

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PDF

Imagine you are expert scribe generator. Make a lecture scribe suitable for exam preparation for CSE400 Lecture 8:Gaussian Distribution .
When you are writing the scribe, do reasoning step by step through definitions, assumptions, proofs, derivations, and worked example solutions exactly as presented in the provided context. Include this reasoning in the scribe so that the logical flow is clear for revision. Use only the provided lecture slides and relevant textbook excerpts. Don't mention any material, explanations or examples that are not present in the context.
Organize the scribe clearly in a formal academic style.

CSE400 – Fundamentals of Probability in Computing

Lecture 8: Gaussian Distribution

(Based strictly on Lecture Slides: L8_S1_A.pdf)

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1. Continuous Random Variables: Moment Foundations

Before introducing the Gaussian random variable, the lecture revises **moments and central moments** (pages 3–4).

1.1 n-th Order Central Moment

For a continuous random variable (CRV),

$$E[(X - \mu_X)^n] = \int (x - \mu_X)^n f_X(x)\, dx$$

For discrete case,

$$\mathrm{E}[(X-\mu_X)^n] = \sum_k (x_k-\mu_X)^n p_X(x_k)$$

Step-by-step evaluation of low-order central moments

(i) Zeroth Order (n = 0)

$$\mathrm{E}[(X-\mu_X)^0] = \mathrm{E}[1] = 1$$

Since any non-zero quantity raised to power 0 equals 1.

(ii) First Order (n = 1)

$$\mathrm{E}[X-\mu_X] = \mathrm{E}[X] - \mu_X$$

But,

$$\mathrm{E}[X] = \mu_X$$

Thus,

$$\mathrm{E}[X-\mu_X] = \mu_X - \mu_X = 0$$

(iii) Second Order (n = 2): Variance

$$\sigma_X^2 = \mathrm{E}[(X-\mu_X)^2]$$

Expand the square:

$$(X-\mu_X)^2 = X^2 - 2\mu_X X + \mu_X^2$$

Taking expectation:

$$\mathrm{E}[X^2] - 2\mu_X \mathrm{E}[X] + \mu_X^2$$

Since $\mathrm{E}[X] = \mu_X$,

$$\sigma_X^2 = \mathrm{E}[X^2] - \mu_X^2$$

1.2 Higher Order Moments

Skewness (n = 3)

$$C_s = \frac{E[(X - \mu_X)^3]}{\sigma_X^3}$$

- $C_s > 0 \rightarrow$ Right-skewed PDF
- $C_s < 0 \rightarrow$ Left-skewed PDF

Kurtosis (n = 4)

$$C_k = \frac{E[(X - \mu_X)^4]}{\sigma_X^4}$$

Large kurtosis → Sharp peak near mean.

1.3 Linearity of Expectation (Theorem)

For constants a, b:

$$E[aX + b] = aE[X] + b$$

For sum of functions:

$$E\left[\sum_{k=1}^{N} g_k(X)\right] = \sum_{k=1}^{N} E[g_k(X)]$$

Also,

$$E[X] = \int x f_X(x)\, dx$$

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2. Gaussian Random Variable

(Definition begins on page 8)

2.1 Definition

A Gaussian random variable is one whose PDF is:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Notation:

$$X \sim \mathrm{N}(m, \sigma^2)$$

Where:

- m = mean
- σ^2 = variance

(See page 9 figure showing PDF and CDF.)

2.2 Interpretation

From the figure (page 9):

- PDF is symmetric about mean m
- Area under PDF = 1
- CDF increases from 0 to 1

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3. Standard Forms

(Page 10–12)

3.1 Error Function

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

3.2 Complementary Error Function

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt$$

3.3 Φ-function (CDF of Standard Normal)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2} dt$$

This gives left-tail probability.

3.4 Q-function (Gaussian Tail Function)

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_{x}^{\infty} e^{-t^2/2} dt$$

This gives right-tail probability.

Key Identity (Page 12)

$$Q(x) = 1 - \Phi(x)$$

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4. Relation Between Φ-function and Q-function

(Page 14–19)

4.1 Evaluating the CDF of Gaussian RV

Start with:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right) dy$$

Standardization Step

Let:

$$t = \frac{y-m}{\sigma}$$

Then:

$$F_X(x) = \int_{-\infty}^{\frac{x-m}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Thus,

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

4.2 Tail Probability

$$P(X > x) = \int_x^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-m)^2/2\sigma^2} dy$$

After substitution:

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

4.3 Key Identities (Page 16)

$$Q(x) = 1 - \Phi(x)$$

$$F_X(x) = 1 - Q\left(\frac{x-m}{\sigma}\right)$$

Important Remarks (Page 17–18)

- To evaluate CDF of Gaussian RV → evaluate Φ-function
- Q-function is more natural for probabilities of form $P(X > x)$

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5. Worked Example

(Page 20–21)

Given PDF

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Step 1: Compare with Gaussian Form

General form:

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Match terms:

$$m = -3$$

$$2\sigma^2 = 8 \Rightarrow \sigma^2 = 4 \Rightarrow \sigma = 2$$

Thus,

$$X \sim \mathrm{N}(-3, 4)$$

Gaussian Relations

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

(i) $P(X \leqslant 0)$

Standardize:

$$\frac{0-(-3)}{2} = \frac{3}{2}$$

$$P(X \leqslant 0) = \Phi\left(\frac{3}{2}\right)$$

(ii) $P(X > 4)$

$$\frac{4-(-3)}{2} = \frac{7}{2}$$

$$P(X > 4) = Q\left(\frac{7}{2}\right)$$

(iii) $P(|X+3| < 2)$

Rewrite:

$$-2 < X + 3 < 2$$

$$-5 < X < -1$$

Thus,

$$P(-5 < X < -1) = F_X(-1) - F_X(-5)$$

Convert using Φ:

$$= \Phi\left(\frac{-1+3}{2}\right) - \Phi\left(\frac{-5+3}{2}\right)$$

(iv) $P(|X-2| > 1)$

Rewrite:

$$|X-2| > 1$$

Two regions:

$$X < 1 \quad \text{or} \quad X > 3$$

Thus,

$$P(X < 1) + P(X > 3)$$

Use Φ for left-tail and Q for right-tail.

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6. Summary for Exam Revision

1. Gaussian PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

2. Standardization:

$$Z = \frac{X-m}{\sigma}$$

3. CDF:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

4. Tail Probability:

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

5. Identity:

$$Q(x) = 1 - \Phi(x)$$

This completes the structured lecture scribe for **CSE400 Lecture 8: Gaussian Distribution**, prepared strictly from the provided lecture slides .

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CSE400 – Fundamentals of Probability in Computing

Lecture 8: Gaussian Distribution

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1. Preliminaries: Moments of Continuous Random Variables

(From pages 3–5)

Before introducing the Gaussian random variable, the lecture develops the concept of **moments and central moments**, which are fundamental in describing distributions.

1.1 n-th Order Central Moment

For a continuous random variable X with mean μ_X:

$$\mathrm{E}[(X - \mu_X)^n] = \int (x - \mu_X)^n f_X(x)\, dx$$

For the discrete case:

$$\mathrm{E}[(X - \mu_X)^n] = \sum_k (x_k - \mu_X)^n p_X(x_k)$$

1.2 Evaluation for Specific Orders

(i) Zeroth Order (n = 0)

$$\mathrm{E}[(X - \mu_X)^0] = \mathrm{E}[1] = 1$$

(ii) First Order (n = 1)

$$\mathrm{E}[X - \mu_X] = \mathrm{E}[X] - \mu_X$$

Since by definition $\mathrm{E}[X] = \mu_X$,

$$\mathrm{E}[X - \mu_X] = 0$$

(iii) Second Order (n = 2): Variance

$$\sigma_X^2 = \mathrm{E}[(X - \mu_X)^2]$$

Expand:

$$(X - \mu_X)^2 = X^2 - 2\mu_X X + \mu_X^2$$

Taking expectation:

$$\mathrm{E}[X^2] - 2\mu_X \mathrm{E}[X] + \mu_X^2$$

Since $\mathrm{E}[X] = \mu_X$,

$$\sigma_X^2 = \mathrm{E}[X^2] - \mu_X^2$$

1.3 Higher Order Central Moments

Skewness (n = 3)

$$C_s = \frac{\mathrm{E}[(X - \mu_X)^3]}{\sigma_X^3}$$

- Positive → Right-skewed PDF
- Negative → Left-skewed PDF

(As illustrated graphically on page 4.)

Kurtosis (n = 4)

$$C_k = \frac{\mathrm{E}[(X - \mu_X)^4]}{\sigma_X^4}$$

Large value indicates a sharp peak near the mean (page 4).

1.4 Linearity of Expectation (Theorem)

(From page 5)

For constants a, b:

$$\mathrm{E}[aX + b] = a\mathrm{E}[X] + b$$

For sum of functions:

$$E\left[\sum_{k=1}^{N} g_k(X)\right] = \sum_{k=1}^{N} E[g_k(X)]$$

Also:

$$E[X] = \int x f_X(x)\, dx$$

---

2. Gaussian Random Variable

(Definition begins page 8–9)

2.1 Definition

A Gaussian random variable is one whose PDF is:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Notation:

$$X \sim \mathrm{N}(m, \sigma^2)$$

Where:

- m = mean
- σ^2 = variance

(Page 9 shows PDF and CDF plots for $m = 3, \sigma = 2$.)

2.2 Observations from the Figure (Page 9)

- The PDF is symmetric about m.
- Total area under PDF = 1.
- CDF increases monotonically from 0 to 1.

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3. Standard Forms

(Pages 10–12)

3.1 Error Function

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

3.2 Complementary Error Function

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt$$

3.3 Φ-function (CDF of Standard Normal)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt$$

This gives **left-tail probability**.

3.4 Q-function (Gaussian Tail Function)

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt$$

This gives **right-tail probability**.

Key Identity (Page 12)

$$Q(x) = 1 - \Phi(x)$$

---

4. Relation Between Φ-function and Q-function

(Pages 14–19)

4.1 Evaluating the CDF of a Gaussian RV

Start with:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right) dy$$

Standardization

Let:

$$t = \frac{y-m}{\sigma}$$

Then:

$$F_X(x) = \int_{-\infty}^{\frac{x-m}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Thus,

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

4.2 Evaluating Tail Probability

$$P(X > x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-m)^2/2\sigma^2} dy$$

After substitution:

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

4.3 Key Identities (Page 16)

$$Q(x) = 1 - \Phi(x)$$

$$F_X(x) = 1 - Q\left(\frac{x-m}{\sigma}\right)$$

Important Remarks (Pages 17–18)

- To evaluate CDF of Gaussian RV → evaluate Φ-function.
- Q-function is more natural for probabilities of form $P(X > x)$.
- Page 19 visually shows left-tail area as Φ(x) and right-tail area as Q(x).

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5. Worked Example

(Pages 20–21)

Given PDF

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Step 1: Identify Parameters

Compare with:

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Matching terms:

$$m = -3$$

$$2\sigma^2 = 8 \Rightarrow \sigma^2 = 4 \Rightarrow \sigma = 2$$

Thus,

$$X \sim \mathrm{N}(-3, 4)$$

Gaussian Relations

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

(i) $P(X \leqslant 0)$

Standardize:

$$\frac{0-(-3)}{2} = \frac{3}{2}$$

$$P(X \leqslant 0) = \Phi\left(\frac{3}{2}\right)$$

(ii) $P(X > 4)$

$$\frac{4-(-3)}{2} = \frac{7}{2}$$

$$P(X > 4) = Q\left(\frac{7}{2}\right)$$

(iii) $P(|X+3| < 2)$

Rewrite:

$$-2 < X + 3 < 2$$

$$-5 < X < -1$$

Thus:

$$P(-5 < X < -1) = F_X(-1) - F_X(-5)$$

Convert:

$$= \Phi\left(\frac{-1+3}{2}\right) - \Phi\left(\frac{-5+3}{2}\right)$$

(iv) $P(|X-2| > 1)$

$$|X-2| > 1$$

Equivalent to:

$$X < 1 \quad \text{or} \quad X > 3$$

Thus:

$$P(X < 1) + P(X > 3)$$

Use Φ for left-tail and Q for right-tail.

---

6. Final Exam-Oriented Summary

1. Gaussian PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

2. Standardization:

$$Z = \frac{X - m}{\sigma}$$

3. CDF:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

4. Tail Probability:

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

5. Identity:

$$Q(x) = 1 - \Phi(x)$$

This completes the structured lecture scribe for **CSE400 Lecture 8: Gaussian Distribution**, prepared strictly from the provided slides .

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Lecture 8: Gaussian Distribution

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1. Foundations: Moments of a Continuous Random Variable

(Developed on pages 3–5)

The lecture begins by revisiting **moments and central moments**, since Gaussian properties are understood through mean and variance.

1.1 n-th Order Central Moment

For a continuous random variable X with mean μ_X:

$$E[(X - \mu_X)^n] = \int (x - \mu_X)^n f_X(x)\, dx$$

For the discrete case:

$$E[(X - \mu_X)^n] = \sum_k (x_k - \mu_X)^n p_X(x_k)$$

1.2 Evaluation for Specific Orders

(i) Zeroth Order (n = 0)

$$E[(X - \mu_X)^0] = E[1] = 1$$

(ii) First Order (n = 1)

$$E[X - \mu_X] = E[X] - \mu_X$$

Since by definition:

$$E[X] = \mu_X$$

Hence:

$$E[X - \mu_X] = 0$$

(iii) Second Order (n = 2): Variance

$$\sigma_X^2 = E[(X - \mu_X)^2]$$

Expand:

$$(X - \mu_X)^2 = X^2 - 2\mu_X X + \mu_X^2$$

Taking expectation:

$$E[X^2] - 2\mu_X E[X] + \mu_X^2$$

Since $E[X] = \mu_X$,

$$\sigma_X^2 = E[X^2] - \mu_X^2$$

1.3 Higher Order Central Moments

Skewness (n = 3)

$$C_s = \frac{E[(X - \mu_X)^3]}{\sigma_X^3}$$

- Positive → Right-skewed PDF

- Negative → Left-skewed PDF

(Page 4 shows graphical interpretation.)

Kurtosis (n = 4)

$$C_k = \frac{\mathrm{E}[(X - \mu_X)^4]}{\sigma_X^4}$$

Large kurtosis indicates a large peak near the mean.

1.4 Linearity of Expectation

(Page 5)

For constants a, b:

$$\mathrm{E}[aX + b] = a\mathrm{E}[X] + b$$

For sum of functions:

$$\mathrm{E}\left[\sum_{k=1}^{N} g_k(X)\right] = \sum_{k=1}^{N} \mathrm{E}[g_k(X)]$$

Also:

$$\mathrm{E}[X] = \int x f_X(x)\, dx$$

---

2. Gaussian Random Variable

(Definition on pages 8–9)

2.1 Definition

A Gaussian random variable is one whose PDF is:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - m)^2}{2\sigma^2}\right)$$

Notation:

$$X \sim \mathrm{N}(m, \sigma^2)$$

Where:

- m = mean
- σ^2 = variance

2.2 Graphical Interpretation (Page 9)

The slide shows:

- PDF symmetric about mean m
- Total area under curve = 1
- Corresponding CDF increases from 0 to 1

---

3. Standard Forms

(Pages 10–12)

3.1 Error Function

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

3.2 Complementary Error Function

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt$$

3.3 Φ-function (CDF of Standard Normal)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt$$

This represents **left-tail probability**.

3.4 Q-function (Gaussian Tail Function)

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_{x}^{\infty} e^{-t^2/2} dt$$

This represents **right-tail probability**.

Key Identity (Page 12)

$$Q(x) = 1 - \Phi(x)$$

---

4. Relation Between Φ and Q

(Pages 14–19)

4.1 Evaluating the CDF

Start from Gaussian CDF definition:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right) dy$$

Standardization Step

Let:

$$t = \frac{y-m}{\sigma}$$

Then:

$$F_X(x) = \int_{-\infty}^{\frac{x-m}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Thus:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

4.2 Tail Probability

$$P(X > x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-m)^2/2\sigma^2} dy$$

After substitution:

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

4.3 Key Identities (Page 16)

$$Q(x) = 1 - \Phi(x)$$

$$F_X(x) = 1 - Q\left(\frac{x-m}{\sigma}\right)$$

Important Observations (Pages 17–19)

- To evaluate CDF → use Φ-function.
- For probabilities of form $P(X > x)$ → Q-function is more natural.
- Page 19 visually shows left-tail area (Φ) and right-tail area (Q).

---

5. Worked Example

(Pages 20–21)

Given

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Step 1: Identify Parameters

Compare with:

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Matching terms:

$$m = -3$$

$$2\sigma^2 = 8 \Rightarrow \sigma^2 = 4 \Rightarrow \sigma = 2$$

Thus:

$$X \sim \mathrm{N}(-3, 4)$$

Gaussian Relations

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

(i) $P(X \leqslant 0)$

$$\frac{0-(-3)}{2} = \frac{3}{2}$$

$$P(X \leqslant 0) = \Phi\left(\frac{3}{2}\right)$$

(ii) $P(X > 4)$

$$\frac{4-(-3)}{2} = \frac{7}{2}$$

$$P(X > 4) = Q\left(\frac{7}{2}\right)$$

(iii) $P(\mid X+3 \mid < 2)$

$$-2 < X + 3 < 2$$

$$-5 < X < -1$$

$$P(-5 < X < -1) = F_X(-1) - F_X(-5)$$

$$= \Phi\left(\frac{-1+3}{2}\right) - \Phi\left(\frac{-5+3}{2}\right)$$

(iv) $P(|X-2| > 1)$

$$|X-2| > 1$$

$$X < 1 \quad \text{or} \quad X > 3$$

$$P(X<1) + P(X>3)$$

Use Φ for left-tail and Q for right-tail.

---

6. Exam-Focused Summary

1. Gaussian PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

2. Standardization:

$$Z = \frac{X-m}{\sigma}$$

3. CDF:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

4. Tail Probability:

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

5. Identity:

$$Q(x) = 1 - \Phi(x)$$

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